

How to Get Smarter With AI

Show Slide 1 - How to Get Smarter With AI

About Me

Hi, I'm Simon, Lucy's Dad. I'm a software engineer and I build things with AI every day. Let's talk about AI!

What Do You Already Know? (2 min)

Show of hands — quick and energetic:

Who has heard of AI before? Who has actually used it — ChatGPT, Siri, Google, anything? Who thinks they know what AI stands for? (*Artificial Intelligence* — confirm if it comes up, don't dwell)

Popcorn around the room — take 4–5 fast responses:

Give me a few words about what you think AI is, or what does it do?

Don't correct or expand yet — just collect and affirm. You'll get answers like: *writes things, answers questions, makes pictures, cheats on homework*. All fine.

Then reframe:

You already know a lot. Today we're not here to talk about what AI can do — you clearly know it can write stuff, answer questions, make images. Today we're going to talk about how to use it in a way that actually makes YOU smarter. Most people use it wrong. You're going to learn how to use it right.

The Question Changes Everything (6 min)

Before slide 2 on the screen, ask:

Imagine you just finished writing a paragraph for English homework and you wanted AI to help you make it better. What would you actually type into ChatGPT?

Take 2–3 responses — they'll almost certainly be short and direct, like "*Make my writing better.*" Don't critique them.

Most students go even further — they just ask AI to write the whole thing. Let's see what that looks like.

Show Slide 2 - The Question Changes Everything

Let's see what happens if we type the lazy prompt: "Write me a short paragraph about my weekend."

We would get an AI Response like: "This weekend was a blast! I went to the park with my friends and we ran around for hours. Sunday was all about relaxing, I watched a movie and fell asleep on the couch."

Whose writing is this?

Get responses and/or prompt:

AI did the writing, correct?

Did you get better at writing, or did AI do the work for you?

Get responses/engagement then read the smart prompt and response aloud. The difference speaks for itself.

Here's the move that changes everything, the smart prompt. "Here's a paragraph I wrote for my English homework about my weekend: 'My dog is fun. I threw a ball to him at the park. We went to the zoo on Sunday. I fell asleep on the couch.' I tried to describe what my weekend was like but I'm not sure if it's interesting enough. Can you give me tips to improve this without rewriting it for me?"

Read AI Response:

Great start! Here are a few tips: - Add feelings: instead of "fun," tell us how it felt. Were you laughing? Was your dog going crazy for the ball? Add one detail about the zoo: what did you see there? One animal would make it come alive. Show the tired feeling: instead of "fell asleep," why did you fall asleep? Was it a big, busy day? Pick just one of these and try adding it to your paragraph!"

Both prompts wanted help with writing — so what's different about the second one?

It shows your actual work, explains what you were trying to do, and asks for tips instead of a rewrite

Why does sharing your work and your goal make such a big difference?"

Golden rule: "AI can't help you improve if it doesn't see what you've already done. Show your work first — and tell it what kind of help you want."

The Power Prompts (8 min)

Show Slide 3 - Power Prompts

Does anyone know what an AI Model is?

Wait for some answers, maybe no one knows.

ChatGPT is an example of an AI model — think of it as one AI brain that was trained on huge amounts of text and information like a human expert. But ChatGPT isn't the only one. Gemini is Google's AI model, Copilot is Microsoft's AI, and Claude AI is made by a company called Anthropic. Each was trained differently, so each has different strengths just like human experts.

Can you think of a reason you'd go to an English teacher expert instead of a science teacher expert, even if both are great?

Wait for some answers

Different experts for different jobs. Same idea with AI models. An English teacher can help me with language, a science teacher could help you understand what a formula is for.

OK, let's talk about prompts! Does anyone know what a prompt is?

Wait for some answers

Yes, it is something you type or say to the AI model

The three prompts you're about to learn, remembering that a prompt is just something you type or say to the AI, will work on any of these models — but the answer you get back might be different depending on which model you use. Part of getting smarter with AI is knowing which model fits your task, and noticing when a different AI model might do better.

Today we're just going to focus on one model, ChatGPT to learn about the power prompts.

I will explain each prompt in turn and why it works. After that we will all have a chance to do these ourselves. So, here are the three prompts.

The Gap Finder, finds a gap in my knowledge. The prompt goes like this: "I think [X] because [reason]. What am I missing or getting wrong?"

The Improver prompt helps us get better at a task. The prompt goes like this: "Here's my work: [paste it]. How could I improve it? Don't rewrite it — just tell me what to change and why."

The Quiz Master prompt helps us know we understand what we learned. It goes like this: "Explain [topic] to me simply. Then give me 3 questions to test if I actually understood it."

So I'll give you some examples

For Gap Finder. I type in what I think: "I think dingoes are just wild dogs that escaped into the bush. What am I missing or getting wrong?"

The AI might respond with this: "Dingoes aren't escaped pet dogs — they're a separate wild animal that came to Australia thousands of years ago. They're more closely related to wolves than to pet dogs."

This works because I gave AI my understanding about dogs and wolves, and it found the exact gap and didn't just give me an explanation of the difference.

OK, let's try the Improver prompt: "Here's my paragraph: 'Dogs are great pets. They are friendly and they like to play. I like dogs.' How could I improve it? Don't rewrite it — just tell me what to change and why."

This works because you stay in control. AI becomes your editor, it doesn't just write it for you!

Always say "don't rewrite it." Without that, AI replaces your work instead of helping you get smarter.

The third prompt is the Quiz Master. For my prompt I wrote: "Explain how dingoes are different from pet dogs in simple terms. Then give me 3 questions to test if I actually understood it."

Dingoes are wild animals that have lived in Australia for thousands of years. Pet dogs were tamed by humans over a long time to live with us. Dingoes are more like wolves — they hunt, live in packs, and don't need humans to survive.

3 Questions to test yourself: Did dingoes escape from people's homes to become wild, or were they always wild animals? Which animal is a dingo more closely related to — a pet dog or a wolf? Could a dingo survive without humans feeding it? Why or why not?

Why this works is that reading an explanation feels like you understand it, but answering questions reveals what you actually know.

Can AI get the answer wrong?

When AI makes a mistake, it sounds like it knows, but when it's wrong we call this a **hallucination**. AI doesn't know it's wrong; it just sounds very sure. That's why the Quiz Master prompt is doubly useful: it catches gaps in *your* thinking, and it gives you something to fact-check against another source.

Before we do some ourselves on our iPads, let's just try to make up some power prompts. Let's ask some questions about our landmarks. Here is mine for the Gap Finder:

Gap Finder: "I think the Eiffel Tower is the tallest building in the world. What am I missing or getting wrong?"

"It was! When it was built in 1889, the Eiffel Tower was the tallest structure in the whole world. But that's the gap — it's not anymore. Today the tallest building is in Dubai, which is more than twice as tall."

So, could the AI knowledge be out of date?

Wait for answers

Does anyone have a question about their landmark that we could ask the Gap Finder, Improver or Quiz Master?

Wait for answers

Prompt Challenge (12 min)

Show Slide 4 - Your Turn Prompt Challenge

Pair up. Open ChatGPT, and let's do three rounds,

Round 1 — Gap Finder (5 min)

If you know your landmark, use that, otherwise, pick any topic you've been learning about maths, science, history, anything. Write one sentence: what you think you know. Then use the Gap Finder prompt.

Walk around, help groups

Who can tell me their prompt and answer. What did AI say you had right? What did it say was missing?

Round 2 — Improver (3 min)

Write a sentence about your landmark if you have one, or anything else if you don't like your dog, a favourite sport, a book, your weekend. Write it yourself first, then use the Improver prompt. Pick one suggestion and fix that one thing.

Walk around, help groups

Did you agree with what AI suggested? Did fixing it make it better?

Round 3 — Quiz Master (3 min)

Pick something you want to understand better. Use the Quiz Master prompt. Try to answer all 3 questions without looking anything up, then ask AI which ones were right.

Did the quiz catch you out? Was there a gap you didn't know you had?

Share back (3 min):

Did it feel like you learned something, or did it feel like AI did the work? Was there a moment where AI showed you something you didn't know you didn't know? Did anyone notice their model giving an answer that seemed off, or missing something?

Wrap-Up (3 min)

Show Slide 5 - What you learned

The takeaways:

1. Don't ask for the answer — show your thinking and ask where you went wrong.
2. The three power prompts: **Gap Finder, Improver, Quiz Master**.
3. AI is most useful when you tell it what you already think. Show your work first.
4. AI can make things up and have hidden unfair assumptions — always think for yourself.
5. Different AI models have different strengths — choosing the right one is a skill.

Things to watch out for:

Hallucinations: AI sometimes makes up facts and states them confidently — this is called a hallucination. It doesn't know it's wrong. If AI tells you something for an assignment, check it against another source. Don't just trust it because it sounds sure.

Bias: AI learned by reading millions of things people wrote online. But not everyone's experiences were included equally. That means AI can sometimes have hidden unfair assumptions — like always picturing a scientist as one particular type of person, or not knowing about things that aren't common in what it read. If something AI says doesn't feel quite right or fair, trust that instinct. Your own thinking matters.

Cut off date: Models learn on data at a point in time, that means anything after that time they don't know. So the model can't answer questions about the weather today, or the footy score on the weekend.

Finally: The smartest people aren't the ones who know the most. They're the ones who are best at finding their own gaps — and best at knowing when to question the answers they're given. You've now got a tool that can help you do both.

Thanks everyone for having me today.